

# LIVE QUIZ PLATFORM

A technical review blueprint detailing complete API endpoints routing, relational schema models, Web Audio synthesizers, and the multi-agent AI quiz generation & verification pipeline.

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The platform is a modular classroom assessment application connecting educators and students in real-time. The portal handles administrative quiz controls, automated question generation, websocket rooms, custom game boards, and deep performance analytics.

## **4 INGESTION WAYS**

PROMPT, PDF, SPEECH  
AUDIO, YOUTUBE URL

## **2 ARENA LOBBIES**

SPRINT MATCH & 3D  
MATCH-UP ARENA

## **ANTI-CHEAT CHECKS**

DUPLICATE LOGINS,  
SERVER-SIDE XP  
CALCULATIONS



Below is the comprehensive catalog of server endpoints, grouped by features and operational access.

## AUTHENTICATION ROUTES (/API/AUTH)

**POST /register** - Admin-restricted user registration. Creates teacher, student, or administrator records in the database with encrypted passwords.

**POST /login** - Authenticates users via username/email. Generates a signed JWT payload containing tokenVersion and user role. **Blocks duplicate student logins.**

**POST /register-public** - Self-registration path for students. Assigns an initial role of `none` prior to workspace setup.

**POST /set-role** - Updates and locks a newly registered user's role (`student` | `teacher`). This can only be executed once.

**GET /me** - Active session validation path. Resolves user authentication status and returns the profile object.

**PUT /change-password** - Updates user password and increments `tokenVersion` in the DB, immediately revoking all previous JWT tokens to force active sessions to re-authenticate.

**POST /forgot-password** - Issues a secure, temporary 10-minute password reset token mapped to the user ID.

## QUIZ & GENERATOR ROUTES (/API/QUIZ)

**POST /generate** - Spawns an asynchronous AI quiz generation task. Returns a unique `taskId` immediately. The parsing handles PDF parsing (`pdf-parse`) and Office formatting.

**GET /generate/status/:taskId** - Status polling endpoint for background quiz generation tasks.

**POST /generate-voice** - Generates questions from uploaded audio recordings using the **\*\*Groq Whisper API\*\*** to transcribe speech to plaintext before calling the generator model.

**POST /create** - Persists a manual or AI-generated quiz to the database.

**GET /my-quizzes** - Fetches all quizzes created by the logged-in educator.

**GET /result/:quizId** - Fetches a student's performance rank and details on completion.

**PATCH /:id/schedule** - Updates start and end dates/times for scheduled homework assignments.

**DELETE /:id** - Cascade deletes a quiz and its questions.

## STUDENT PORTAL ROUTES (/API/STUDENTS)

**GET /gamification** - Retrieves XP, login streak metrics, daily missions, and unlocked perks.

**POST /gamification/init** - Handles streak checks and daily mission resets.

**POST /game-score** - Submits game results from Sprint or Match-Up. **XP is calculated server-side** based on questions, duration, and moves to prevent client-side cheat farming.

**POST /redeem-perk** - Spends student XP to redeem virtual classroom passes, checking monthly limits per ticket.

To guarantee high-quality pedagogical assessments, the system does not just call an LLM directly. It executes a structured **Agentic Quality Assurance Pipeline**.

## INTERACTIVE QA PIPELINE FLOW

### STAGE 1: THE GENERATOR (GOOGLE GEMINI 1.5)

Processes raw plaintext input and generates a draft set of questions, options, and explanations based on the requested count and topic.

### STAGE 2: THE CRITIC AGENT (RULE-BASED EXPERT SYSTEM)

Scores each question from 0 to 100 based on 6 weighted criteria: Correctness (30 pts), Clarity (20 pts), Distractor Quality (15 pts), Explanation Length (10 pts), Difficulty Alignment (15 pts), and Concept Uniqueness (10 pts). It also flags whole-quiz issues like answer-position bias or repeated concepts.

### STAGE 3: EARLY EXIT CHECK

#### OPTIMIZATION

If the average score of all draft questions is already  $\geq 92$ , the pipeline skips refinement completely and returns the draft questions. This saves API usage costs and reduces latency.

### STAGE 4: THE REFINER AGENT (LLAMA 3.1 8B VIA GROQ)

Questions scoring below 95 are refined in parallel:

- **Minor Refinement (85-94)**: Only punctuation, explanation length, and formatting are corrected.
- **Full Refinement (<85)**: The question is rewritten based on the specific feedback issues generated by the Critic.

### STAGE 5: FINAL VALIDATION & DIFF GENERATION

The Critic re-scores the refined questions. The pipeline returns the finalized questions alongside a structured report showing the quality score improvement (e.g., from 78 to 96) and a visual diff of the modified questions.

Live classrooms run on a custom WebSocket state room. Below is the protocol matrix:

| EVENT NAME             | DIRECTION           | TRIGGER DESCRIPTION                                                     |
|------------------------|---------------------|-------------------------------------------------------------------------|
| join_room              | Client →<br>Server  | Student joins the lobby using a PIN code.<br>Registers presence.        |
| reconnectUser          | Client →<br>Server  | Restores student quiz progress from the<br>database after a disconnect. |
| start_quiz             | Teacher →<br>Server | Teacher starts the quiz. Emits the first<br>question to the room.       |
| submit_question_answer | Client →<br>Server  | Submits an answer choice and checks speed<br>against the lobby average. |
| answer_feedback        | Server →<br>Client  | Emits response speed feedback (cyan or amber<br>popups) to the student. |

```
model User {  
  
    id String @id @default(uuid())  
  
    username String @unique  
  
    email String @unique  
  
    password String  
  
    role String // 'student' | 'teacher' | 'admin'  
  
    isOnline Boolean @default(false)  
  
    xp Int @default(0)  
  
    dailyMissions Json?  
  
    unlockedPerks Json?  
  
    createdAt DateTime @default(now())  
  
}
```

```
model Quiz {  
  
    id String @id @default(uuid())  
  
    title String  
  
    topic String  
  
    creatorId String  
  
    questions Question[]  
  
    isLive Boolean @default(false)  
  
    isAssessment Boolean @default(false)  
  
    createdAt DateTime @default(now())  
  
}
```

```
model Question {
```



Use these key bullet points to highlight the technical achievements of the project during your review:

- **Agentic Quiz QA Pipeline:** Emphasize that the platform uses an intelligent Critic-Refiner loop with an *early-exit check* to ensure high question quality while keeping token usage efficient.
- **3-Tier YouTube Ingestion Pipeline:** Discuss the fallback structure (Transcript → Metadata → Whisper Audio) that ensures 95% of video URLs work seamlessly.
- **Browser Stacking Glitch Resolution:** Explain how we resolved Chrome's rounded-corner 3D flattening bug in the Match-Up card game by using explicit transforms, dynamic z-indexing, and opacity cross-fading.
- **Server-Side XP Validation:** Highlight the anti-farming rules that validate gameplay duration and matched pairs before awarding XP, preventing students from cheating.